Topology-Preserving Transform (TPT) State-Variable Filters

Zero-Delay Feedback Filter Discretization & Moog 24dB Cascade Math

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Abstract

This paper details the continuous-time and discretized equations of the active State Variable Filter (`[filter~]` and `[svfilter~]`) implemented using Vadim Zavalishin's Topology-Preserving Transform (TPT) zero-delay feedback method.

1 Continuous-Time Active SVF Equations

The continuous-time 2-pole state-variable filter integrates high-pass $y_{HP}(t)$, band-pass $y_{BP}(t)$, low-pass $y_{LP}(t)$, and notch $y_{\text{notch}}(t)$ outputs simultaneously:

$$y_{HP}(t) = x(t) - R \cdot y_{BP}(t) - y_{LP}(t) \tag{1}$$

$$y_{BP}(t) = \omega_0 \int y_{HP}(t)\,dt \tag{2}$$

$$y_{LP}(t) = \omega_0 \int y_{BP}(t)\,dt \tag{3}$$

$$y_{\text{notch}}(t) = y_{HP}(t) + y_{LP}(t) \tag{4}$$

where $\omega_0 = 2\pi f_c$ is the cutoff angular frequency and $R = \frac{1}{Q}$ is the damping factor.

2 Bilinear Transform Discretization & TPT Structure

Applying trapezoidal integration with pre-warped gain coefficient $g = \tan\left(\frac{\pi f_c}{f_s}\right)$ and resonance damping coefficient $k = \frac{1}{Q}$:

$$v_H(n) = \frac{x(n) - (g+k)s_1(n-1) - s_2(n-1)}{1 + g(g+k)} \tag{5}$$

$$v_B(n) = g \cdot v_H(n) + s_1(n-1) \tag{6}$$

$$v_L(n) = g \cdot v_B(n) + s_2(n-1) \tag{7}$$

State registers $s_1(n)$ and $s_2(n)$ are updated per sample as:

$$s_1(n) = 2v_B(n) - s_1(n-1) \tag{8}$$

$$s_2(n) = 2v_L(n) - s_2(n-1) \tag{9}$$

3 Filter Mode Combinations

The active output signal $y(n)$ is selected according to filter mode:

$$y(n) = \begin{cases} v_L(n), & \text{Low-Pass (LP)} \\ v_H(n), & \text{High-Pass (HP)} \\ v_B(n), & \text{Band-Pass (BP)} \\ v_H(n) + v_L(n), & \text{Notch / Band-Reject} \end{cases} \tag{10}$$

4 Moog 24dB 4-Pole Resonant Lowpass Cascade (`[filter~]`)

For 4-pole 24 dB/octave attenuation, two identical TPT 2-pole SVF stages are cascaded in series:

$$y_{\text{stage1}}(n) = \text{TPT\_SVF}_1\left(x(n),\, f_c,\, Q\right) \tag{11}$$

$$y_{\text{stage2}}(n) = \text{TPT\_SVF}_2\left(y_{\text{stage1}}(n),\, f_c,\, Q\right) \tag{12}$$

This structure maintains amplitude stability and prevents digital cramping near the Nyquist limit $f_s/2$.